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TO WHOM IT MAY CONCERN

This letter is to certify that Mr. Shiva Prasad Anjanappa made significant and original technical contributions to the research project entitled “A Digital Three-Level Space Vector Modulator for High-Frequency Vector Sequence Generation,” published in IEEE Journal of Emerging and Selected Topics in Industrial Electronics, January 2025, PP(99):1–6, DOI:10.1109/JESTIE.2025.3543271.
This work was carried out under the supervision of Dr. Subhransu Satpathy and Dr. Partha Pratim Das, in collaboration with Mr. Anand Kulkarni.

Mr. Anjanappa was responsible for the FPGA (Programmable Logic) design and implementation, where he demonstrated a strong and advanced understanding of digital system architecture. In particular, he designed and validated a robust multi-clock handshaking mechanism to safely coordinate data transfer between two independent finite state machines (FSMs) operating under different clock domains. His work ensured deterministic behavior, data integrity, and metastability-safe operation, which were critical to the correctness of the overall system.

He also demonstrated a solid grasp of pipelined hardware design, carefully structuring the control and data paths to meet timing constraints while maintaining high throughput. Mr. Anjanappa implemented the PWM logic in the FPGA to ensure that the six gate drivers were actuated in a carefully controlled sequence, with an enforced 80-microsecond dead-time gap between specific driver transitions. This timing constraint was critical to preventing unsafe switching conditions and ensuring the reliable and safe operation of the GaN-based 3L-ANPC system.

The Processor System (PS) software was originally developed by Mr. Anand Kulkarni. Mr. Anjanappa thoroughly analyzed this software to understand the control flow and configuration mechanisms and subsequently used this understanding to modify and configure different switching sequences. This hardware–software co-design effort was essential to achieving the experimental results reported in the IEEE publication.

Mr. Anjanappa’s contributions reflect a high level of technical competence in FPGA architecture, FSM design, pipelining, and multi-clock domain synchronization, skills that are typically developed through sustained, research-intensive engagement in this field. His work materially contributed to the successful realization of the system presented in the paper.

Based on his technical depth, originality of contribution, and ability to independently address complex hardware design challenges, Mr. Anjanappa has demonstrated strong and steadily developing research capabilities in FPGA-based digital system design and hardware–software co-design. His work reflects a solid foundation in advanced concepts such as pipelining, finite state machine design, and multi-clock domain coordination, and indicates a clear trajectory toward specialization in this area.
Sincerely,

S. Bhattacharya.

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